(BINARY) RELATIONS

DEF A CLASS R IS A (BINARY) RELATION IF $R \subseteq V \times V$

$x R y$. A SHORTCUT FOR $(x,y) \in R$

m-ARY RELATION IF $R \subseteq V^m$

EXAMPLE MEMBERSHIP RELATION E IS $\{(x,y) ; x \in y\}$

IDENTITY —"— Id IS $\{(x,y) , x = y\}$

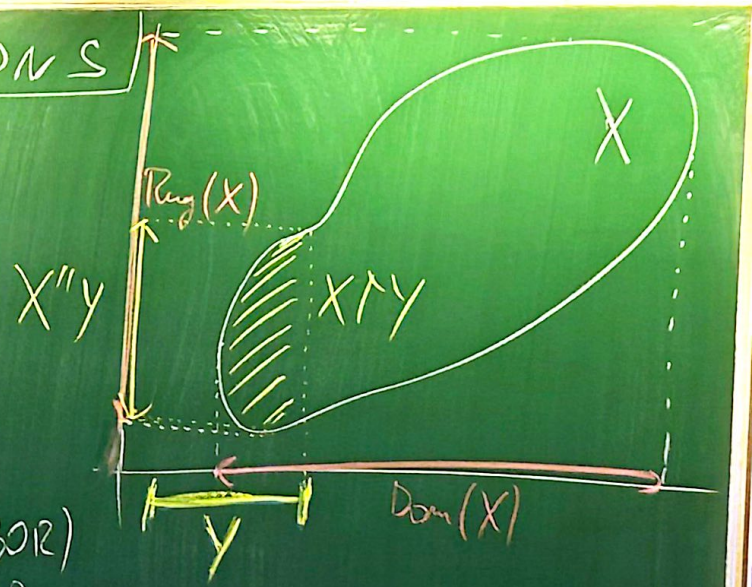
DEF. IF X IS A RELATION (OR AN ARBITRARY CLASS) THEN

$Dom(X)$ IS $\{m , (\exists n)((m,n) \in X\}$ .. THE <u>DOMAIN</u> OF X (DEFINIČNÍ OBOR)

$Rng(X)$ IS $\{n ; (\exists m)((m,n) \in X\}$ .. THE <u>RANGE</u> OF X (OBOR HODNOT)

IF Y IS A CLASS, THEN $X''Y$ (ALSO $X[Y]$) IS $\{z ; (\exists y)(y \in Y \wedge (y,z) \in X\}$ .. IMAGE OF CLASS Y BY CLASS X

$X \restriction Y$ IS $\{(y,z) ; y \in Y \wedge (y,z) \in X\}$ .. RESTRICTION OF CLASS X TO THE CLASS Y (ZÚŽENÍ)
(PARCIALIZATION)



Scanned with CamScanner

$\boxed{\text{LEMMA}}$ IF x IS A SET, Y IS A CLASS, THEN $\mathrm{Dom}(x)$, $\mathrm{Rng}(x)$, $x''Y$, $x\restriction Y$ ARE SETS.

PROOF: CLAIM. $\mathrm{Dom}(x) \subseteq \bigcup(\bigcup x)$:

IF $u \in \mathrm{Dom}(x)$ ... THERE EXISTS v SUCH THAT $(u,v) \in x$

"$u \in \underbrace{\{u\} \in (u,v)}_{\{\{u\},\{u,v\}\}} \in x$" SO $\{u\} \in \bigcup x$ SO $u \in \bigcup(\bigcup x)$

SIMILARLY: $\mathrm{Rng}(x) \subseteq \bigcup(\bigcup x)$: $v \in \mathrm{Rng}$ THEN THERE EXISTS u SUCH THAT $(u,v) \in x$

"$v \in \{u,v\} \in (u,v) \in x$" SO $v \in \bigcup(\bigcup x)$.

$x''Y \subseteq \mathrm{Rng}(x)$, $x\restriction Y \subseteq x$

$\boxed{\text{DEF}}$ R, S RELATIONS THEN R^{-1} IS $\{(u,v) : (v,u) \in R\}$ ... THE RELATION INVERSE TO R

$R \circ S$ IS $\{(u,w) : (\exists v)((u,v) \in R \wedge (v,w) \in S)\}$. THE COMPOSITION OF RELATIONS R, S (SLOŽENÍ)

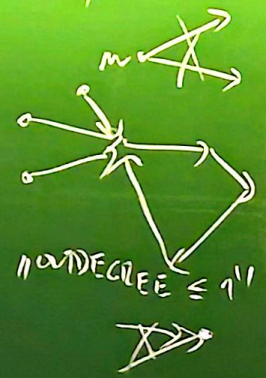
$\boxed{\text{EXERCISE}}$ VERIFY THAT FOR EVERY RELATION R WE HAVE $\mathrm{Id} \circ R = R = R \circ \mathrm{Id}$.

AND THAT FOR EVERY PAIR OF SETS x, y WE HAVE $(x,y) \in E \circ E \iff x \in U_y$

DEF A RELATION F IS A <u>MAPPING</u> (<u>FUNCTION</u>) IF $(\forall u)(\forall v)(\forall w)((\langle u,v\rangle \in F \wedge \langle u,w\rangle \in F) \rightarrow v = w)$

"<u>FOR</u> EVERY $u \in \mathrm{Dom}(F)$ THERE EXISTS EXACTLY ONE SET v SUCH THAT $\langle u,v\rangle \in F$".

WE WRITE $F(u) = v$ $\quad (\langle u,v\rangle \in F \quad , \quad uFv)$



F IS A <u>MAPPING OF A CLASS X</u> <u>TO</u> (INTO) A <u>CLASS Y</u> ... $F : X \rightarrow Y$ IF $\mathrm{Dom}(F) = X$ AND $\mathrm{Rng}(F) \subseteq Y$

F IS A MAPPING OF A CLASS X <u>ONTO</u> (NA) A CLASS Y IF IN ADDITION $\mathrm{Rng}(F) = Y$

F IS <u>INJECTIVE</u> (PROSTÉ) IF F^{-1} IS A MAPPING ... SO $(\forall u)(\forall v)(\forall w)((F(u) = w \wedge F(v) = w) \rightarrow u = v)$.

"EVERY ELEMENT OF $\mathrm{Rng}(F)$ HAS EXACTLY ONE PREIMAGE".

(THEN ALSO F^{-1} IS INJECTIVE).

$$(\exists x \in A)(\overbrace{x = x \rightarrow x \in X}^{\varphi})$$

$\boxed{\text{DEF}}$ (SHORTCUT) A A CLASS, φ A FORMULA, THEN $(\exists x \in A)\varphi$ IS A SHORTCUT FOR $(\exists x)(x \in A \wedge \varphi)$

$(\forall x \in A)\varphi$ — " — $(\forall x)(x \in A \rightarrow \varphi)$

Scanned with CamScanner

NOTATION IMAGE / PREIMAGE OF A CLASS X BY A MAPPING F

$F[X]$ INSTEAD OF $F''X$ $\quad F[X] = \{y ; (\exists x \in X)(y = F(x))\}$

$F^{-1}[X]$ INSTEAD OF $F^{-1}{}''X$ $\quad F^{-1}[X] = \{y ; (\exists x \in X)(x = F(y))\}$

$$a \xrightarrow{f} \mathrm{Rng}(f) \subseteq A$$

DEF A A CLASS, a A SET THEN ${}^{a}A$ IS $\{f ; f : a \to A\}$ .. THE CLASS OF ALL MAPPINGS OF a TO A.

OBSERVATION EVERY MAPPING f OF a TO A IS A SET: $f \subseteq a \times A$, ALSO $f \subseteq a \times \mathrm{Rng}(f)$

- $\mathrm{Rng}(f)$ IS A SET BY REPLACEMENT
- $a \times \mathrm{Rng}(f)$ IS A SET, SO f IS A SET

EXAMPLES: ${}^{\emptyset}y = \{\emptyset\}$

$|y|^{0} = 1$, $0^{|x|}$ $\quad {}^{x}\emptyset$ FOR $x \neq \emptyset$ $= \emptyset$

${}^{B}A$ ..? $F : B \to A$ $\quad$ CAN F BE A SET?

Scanned with CamScanner

/ (ORDERS)

V ON A CLASS A IS

LEMMA 1) FOR ARBITRARY SETS x, y, THE CLASS ${}^{x}y$ IS A SET

2) IF $x \neq \emptyset$, Y IS A PROPER CLASS, THEN ${}^{x}Y$ IS A PROPER CLASS.

PROOF 1) IF $f: x \to y$ THEN $f \subseteq x \times y$, SO $f \in \mathcal{P}(x \times y)$ SO ${}^{x}y \subseteq \mathcal{P}(x \times y)$

$({}^{x}y \subseteq \mathcal{P}(\mathcal{P}(\mathcal{P}(x \cup y))))$

$\in \mathcal{P}(\mathcal{P}(\mathcal{P}(\mathcal{P}(x \cup y))))$

2) FOR $y \in Y$ WE DEFINE A CONSTANT MAPPING $k_y : x \to Y$

SUCH THAT $(\forall m \in x)(k_y(m) = y)$ $\quad (k_y = x \times \{y\})$

- FOR $y \neq y'$ WE HAVE $k_y \neq k_{y'}$ BECAUSE $x \neq \emptyset$

- LET K BE $\{k_y ; y \in Y\}$ .... $K \subseteq {}^{x}Y$. BY CONTRADICTION: IF ${}^{x}Y$ IS A SET, THEN K IS ALSO A SET.

WE DEFINE A MAPPING $F: K \to Y$ AS $F(k_y) = y$. BY REPLACEMENT $\operatorname{Rng} F$ IS A SET, BUT $\operatorname{Rng} F = Y$ SO Y IS A SET - A CONTRADICTION

(ALTERNATIVELY: $\bigcup\bigcup {}^{x}y = x \cup y$ (IF $x \neq \emptyset$ AND $y \neq \emptyset$)

NOTATION IN

$F[X]$ INSTEA

$F^{-1}[X]$ INSTE

DEF A A CLASS, a

OBSERVATION E

- $\operatorname{Rng}(f)$

- $a \times \operatorname{Rng}$

EXAMPLES: ${}^{\emptyset}Y$

$|Y|^{0} \sim 0^{|x|}$ $\quad {}^{x}\emptyset$ FOR

Scanned with CamScanner